SimpleMail, Documentation.

Group Details:

Member 1: Sumedh Jamadagni, 24CSB0A76

Member 2: Tarun Nagarajan 24CSB0A79

Purpose and architecture

This document covers the components and behavior required by the Computer Networking Programming Assignment. SimpleMail consists of `smserver`, `smspooler`, and `smclient`. The client and server use TCP; the server and spooler exchange complete messages through a shared-memory ring. Only the spooler creates and writes mail files. The authenticated server reads or deletes files for SMP mailbox operations.

The server uses nonblocking sockets and `poll()` to serve up to 128 clients on one configurable port. Each connection owns an input buffer, output buffer, protocol state, nonce, and bounded message staging record. Partial TCP reads are retained until a complete command or chunk arrives; partial writes remain queued. A waiting sender does not block other clients.

Build, startup and user registry

Build on Ubuntu/Linux with a C11 compiler, Make, POSIX shared memory, sockets and pthread support:

```
make clean
make
```

The Makefile builds all three programs with `-Wall -Wextra -Wpedantic` and links pthread/real-time support. Start the following in separate terminals, from the same project directory:

```
./smspooler users.txt
./smserver 2525 users.txt
./smclient 127.0.0.1 2525
```

Both daemons load the user registry at startup. Use one username/password pair per line, separated by a space. Usernames are ASCII letters, at most 20 characters, and are compared without case; mailbox paths use lowercase names. Passwords are case-sensitive ASCII alphanumeric strings, at most 30 characters. Duplicate usernames and invalid or missing registries cause startup failure. The parser also permits blank lines and comment lines beginning with #.

The sample users are `alice password1`, `bob secret2`, and `carol hunter7`. The implementation supports 128 registry entries. Passwords in this registry and the required DJB2 challenge mechanism follow the assignment's format.

Start the spooler before the server; the server waits up to five seconds for its shared object. A second live spooler is rejected. After a crash, a replacement spooler can remove a stale shared object whose owner is no longer alive. Stop the server before the spooler; the server must not continue using a queue without its consumer.

Shared-memory spool and synchronization

The POSIX object `/simplemail_v1_ring` contains eight fixed-capacity message slots. Each record holds the sender display name, accepted recipients and count, subject, body length, a 65,536-byte body buffer, and queue ticket. The queue header contains the owner PID, initialization magic, head/tail/count, ticket counter and a dequeue generation counter.

A process-shared pthread mutex protects record publication, indexes and counters. Process-shared `not_empty` and `not_full` conditions coordinate queue availability. These implement the assignment's permitted mutex/condition-variable alternative to unnamed semaphores. The spooler waits for records and releases queue capacity after dequeue. The frontend attempts insertion without waiting in the event loop; a completed message remains in its connection state until a slot is available.

The server assigns increasing queue tickets while the shared queue exists. The spooler validates each queued recipient and writes a copy to every accepted recipient. A fresh spooler run resets the queue ticket sequence; mailbox IDs are separate, durable counters. Queue contents are volatile, so `OK Queued` confirms acceptance into shared memory rather than completed disk persistence.

Mailbox files and IDs

At startup the spooler creates `mailboxes/` and one directory for every registered user. Each mail is a separate `<id>.txt` file. The stored format is:

```
From: <display name>
To: <comma-separated accepted recipients>
Subject: <subject or (no subject)>
Date: <YYYY-MM-DD HH:MM:SS>
---

<body>
```

Each mailbox's `.next-id` metadata starts at 1. The spooler reserves the next ID using a temporary metadata file, flushes and synchronizes it, renames it, and synchronizes the directory before creating the mail. A crash may skip an ID, but deletion and normal restart do not reuse it. A newly created empty mailbox can initialize metadata; an existing mailbox with missing or malformed metadata fails closed. The spooler does not reconstruct deletion history from remaining files.

Connection handshake and SMTP2

A new connection receives `WELCOME SimpleMail v1.0`. `MODE SEND` or `MODE RECV` receives `OK`; RECV then receives its authentication challenge. An invalid mode receives `ERR Unknown mode` and closes. A connection that sends no mode is closed after 30 seconds.

SMTP2 commands and responses end with CRLF. Command lines are limited to 512 bytes including CRLF. Sending requires no authentication; FROM carries a display name, rather than a registered account. The implementation permits 128-byte display names, up to 16 accepted recipients, and printable header text.

Command or state	Required behavior
FROM <display name>	Start a transaction; reply `OK Sender accepted.`
TO <username>	Accept registered recipients with `OK Recipient accepted.`; unknown users receive `ERR No such user.`
SUB <subject>	Require at least one accepted recipient; reply `OK Subject accepted.`; empty text is stored as `(no subject)`.
BODY_STREAM	Enter chunked body input; initially reply `OK Window 2048`.
CHUNK <byte_count>	Read exactly that many raw bytes after the header's CRLF; acknowledge the remaining window.
END_BODY	Queue the complete message, then reply `OK Queued <ticket> to N mailboxes`.
BODY	Reply `OK Send body, end with CRLF.CRLF.` and accept legacy dot-terminated input.
QUIT	Reply `BYE` and close after completing the transaction.

Out-of-sequence commands receive `ERR Bad sequence`. After a queued message, another FROM can start the next transaction on the same sending connection. Unknown recipients do not prevent subsequent valid TO commands. The client must receive at least one acceptance before proceeding to SUB.

In BODY_STREAM, byte counts frame the payload; raw data may contain line breaks and may arrive in several TCP reads. No extra separator is inserted between a chunk's raw bytes and the next command. In BODY, a line containing only `.` terminates input, including an empty body. A sender adds a dot to a body line that already starts with one; the server removes this extra dot. The combined body limit is 65,536 bytes in either mode; exceeding it returns `ERR Body too large.`.

Byte credits and backpressure

The byte window and the ring's message-slot count are different resources. Each complete chunk consumes credits and receives `OK Window <remaining>`. Consumed bytes move into the connection's bounded body record and become pending for credit restoration. A later `WINDOW_UPDATE <additional_credits>` restores those bytes when the spool queue has available capacity. This allows a first large upload to progress without an unrelated prior message.

When the ring is full, credit restoration waits. A sender that consumes its initial 2,048 bytes can therefore receive `OK Window 0` and must pause. The poll loop continues serving other clients. Once the spooler drains the ring, credit updates and waiting queue confirmations resume. Sending a chunk larger than the advertised balance, including any positive chunk at zero credits, returns `ERR Window overflow` and closes the connection. Each body remains bounded independently at 65,536 bytes.

SMP authentication and mailbox operations

After RECV selection, the server sends `AUTH REQUIRED <nonce>`, using eight random alphanumeric characters. The client hashes password followed by nonce, with no separator, using the assignment's DJB2 algorithm, and sends `AUTH <username> <unsigned-decimal hash>`. Successful authentication returns `OK Welcome <canonical username>`. Failed attempts receive `ERR Authentication failed`; the third receives `ERR Too many failures` and closes the connection.

Authenticated command	Response
LIST	`OK <count> messages`, then tab-separated ID, sender display name, subject and date rows; end with a single `.` line.
READ <id>	`OK`, followed by the mail headers/body, with dot-stuffing for body lines beginning with `.`; end with a single `.` line.
DELETE <id>	Remove that user's file and reply `OK Deleted.`; retain ID history.
COUNT	`OK <count>.`
QUIT	`BYE`, then close.
Unrecognized command	`ERR Unknown command.`

Missing READ/DELETE targets receive an error and leave the mailbox unchanged. Authentication fixes the mailbox identity for the session; IDs are interpreted within that user's directory. Client and server must use the same unsigned-long ABI width for DJB2. Command/header text is ASCII-compatible; streamed bodies are raw bytes. Dates and daemon logs use the Ubuntu host's local timezone.

Interactive client and event logs

`smclient <server_ip> <port>` provides the required numbered menu: 1 sends, 2 checks a mailbox, and 3 quits. Sending prompts for display name, recipients, subject and body; a blank recipient finishes the list, and a single-dot body line finishes terminal entry. The client handles chunks, credits and asynchronous updates. Mailbox commands provide list, read, delete, count and logout. Logout returns to the main menu using a fresh connection for the next mode. Retrieval removes transport dot-stuffing before displaying the body.

Both daemons write significant events to stdout with `[YYYY-MM-DD HH:MM:SS]` prefixes. Server events cover connection, mode, authentication, queue insertion, mailbox access and disconnection. Spooler events cover dequeue and per-recipient delivery or failure. These logs distinguish queue acceptance from later file delivery.

Verification and troubleshooting

The current source builds without warnings on Ubuntu 24.04.3 LTS, x86_64, GCC 13.3.0. Ten integration tests cover protocol, storage, concurrency and failure handling. Further scenario runs below force the assignment's boundary and recovery conditions, including an actual 30-second idle timeout, a stopped consumer, zero credits, and twenty simultaneous senders. The real interactive client is also exercised. Test scripts and capture drivers remain local; captured output is included in the documentation.

If the shared queue cannot be opened, start the spooler first. If binding fails, use a free port. For registry errors, correct the characters, lengths or duplicate name. For authentication failure, check password case and hash the password before the current nonce. For an unsafe ID counter, preserve the mailbox and restore valid metadata instead of resetting the number. For delayed sending with a full queue, inspect the spooler and delivery logs; queue confirmations and credit restoration wait for consumer capacity.

Ubuntu execution output

Build and regression results (1/2)

Ubuntu | Build and regression results (1/2)

```
$ lsb_release -ds
Ubuntu 24.04.3 LTS
$ uname -m
x86_64
$ gcc --version | head -1
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
$ make clean
rm -f smserver smspooler smclient *.o
$ make
cc -D_POSIX_C_SOURCE=200809L -std=c11 -O2 -Wall -Wextra -Wpedantic -c -o smserver.o smserver.c
cc -D_POSIX_C_SOURCE=200809L -std=c11 -O2 -Wall -Wextra -Wpedantic -c -o ipc.o ipc.c
cc -D_POSIX_C_SOURCE=200809L -std=c11 -O2 -Wall -Wextra -Wpedantic -c -o common.o common.c
cc  -o smserver smserver.o ipc.o common.o -pthread -lrt
cc -D_POSIX_C_SOURCE=200809L -std=c11 -O2 -Wall -Wextra -Wpedantic -c -o smspooler.o smspooler.c
cc  -o smspooler smspooler.o ipc.o common.o -pthread -lrt
cc -D_POSIX_C_SOURCE=200809L -std=c11 -O2 -Wall -Wextra -Wpedantic -c -o smclient.o smclient.c
cc  -o smclient smclient.o common.o -pthread -lrt
$ python3 tests/integration.py
test_auth_success_casefold_and_three_failures
(__main__.SimpleMailIntegration.test_auth_success_casefold_and_three_failures) ... ok
test_concurrent_sessions_and_command_order
(__main__.SimpleMailIntegration.test_concurrent_sessions_and_command_order) ... ok
test_corrupt_or_missing_id_metadata_fails_closed_after_deletion
```

Build and regression results (2/2)

Ubuntu | Build and regression results (2/2)

```
(__main__.SimpleMailIntegration.test_corrupt_or_missing_id_metadata_fails_closed_after_deletion) ...
 ok
test_empty_subject_legacy_body_and_restart_id_nonreuse
(__main__.SimpleMailIntegration.test_empty_subject_legacy_body_and_restart_id_nonreuse) ... ok
test_fragmented_send_multi_recipient_list_read_delete_and_ids
(__main__.SimpleMailIntegration.test_fragmented_send_multi_recipient_list_read_delete_and_ids) ...
ok
test_max_body_and_over_credit_rejection
(__main__.SimpleMailIntegration.test_max_body_and_over_credit_rejection) ... ok
test_queue_full_blocks_enqueue_then_recovers
(__main__.SimpleMailIntegration.test_queue_full_blocks_enqueue_then_recovers) ... ok
test_spooler_fails_closed_on_corrupt_id_metadata
(__main__.SimpleMailIntegration.test_spooler_fails_closed_on_corrupt_id_metadata) ... ok
test_spooler_refuses_missing_history_for_existing_mailbox
(__main__.SimpleMailIntegration.test_spooler_refuses_missing_history_for_existing_mailbox) ... ok
test_startup_rejects_bad_registry_and_missing_queue
(__main__.SimpleMailIntegration.test_startup_rejects_bad_registry_and_missing_queue) ... ok

----------------------------------------------------------------------
Ran 10 tests in 21.878s

OK
```

Registry errors and startup order

Ubuntu | Registry errors and startup order

```
Ubuntu 24.04.3 LTS | Startup checks and mailbox creation

$ ./smspooler missing.txt
smspooler: cannot open registry 'missing.txt': No such file or directory
exit status: 1
$ ./smserver 2525 missing.txt
smserver: cannot open registry 'missing.txt': No such file or directory
exit status: 1
$ ./smspooler bad.txt
smspooler: invalid registry entry on line 1
exit status: 1
$ ./smserver 2525 users.txt  # spooler is absent
smserver: cannot open spooler shared queue: No such file or directory
$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 34485 users.txt
$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 34485 users.txt
[2026-10-29 10:38:21] spooler: ready; 3 registered users, queue capacity 8
[2026-10-29 10:38:21] server: listening on port 34485 with 3 users
mailboxes/: alice, bob, carol

RESULT: PASS
```

Unknown mode and idle timeout

Ubuntu | Unknown mode and idle timeout

```
Ubuntu 24.04.3 LTS | Unknown mode and 30-second idle timeout

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 50793 users.txt
TCP connection: 127.0.0.1:50793
S> WELCOME SimpleMail v1.0
C> MODE INVALID
S> ERR Unknown mode
[connection closed by server]
TCP connection: 127.0.0.1:50793
S> WELCOME SimpleMail v1.0
[no MODE sent; waiting for the actual 30-second deadline]
[connection closed by server]
observed idle interval: 30.05 seconds

RESULT: PASS
```

Sequence and recipient validation

Ubuntu | Sequence and recipient validation

```
Ubuntu 24.04.3 LTS | Recipients, default subject and dot-stuffed legacy mail

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 59679 users.txt
TCP connection: 127.0.0.1:59679
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> SUB too early
S> ERR Bad sequence
C> FROM Sumedh Jamadagni
S> OK Sender accepted.
C> SUB missing recipient
S> ERR Bad sequence
C> TO ghost
S> ERR No such user.
C> TO BOB
S> OK Recipient accepted.
C> TO carol
S> OK Recipient accepted.
C> SUB
S> OK Subject accepted.
C> BODY
S> OK Send body, end with CRLF.CRLF.
C> legacy text
C> ..leading dot
C> .
S> OK Queued 1 to 2 mailboxes
bob: 1 delivered message; carol: 1 delivered message
```

Default subject and dot stuffing (1/2)

Ubuntu | Default subject and dot stuffing (1/2)

```
TCP connection: 127.0.0.1:59679
S> WELCOME SimpleMail v1.0
C> MODE RECV
S> OK
S> AUTH REQUIRED 3IwFL33L
C> AUTH bob 13450275616599946516
S> OK Welcome bob
C> READ 1
S> OK
S> From: Sumedh Jamadagni
S> To: bob,carol
S> Subject: (no subject)
S> Date: 2026-10-29 10:38:53
S> ---
S>
S> legacy text
S> ..leading dot
S> .
TCP connection: 127.0.0.1:59679
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
```

Default subject and dot stuffing (2/2)

Ubuntu | Default subject and dot stuffing (2/2)

```
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
TCP connection: 127.0.0.1:59679
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Empty Body
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB empty
S> OK Subject accepted.
C> BODY
S> OK Send body, end with CRLF.CRLF.
C> .  [empty body]
S> OK Queued 2 to 1 mailboxes

RESULT: PASS
```

Authentication and mailbox errors

Ubuntu | Authentication and mailbox errors

```
Ubuntu 24.04.3 LTS | Authentication limit and mailbox error cases

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 46171 users.txt
TCP connection: 127.0.0.1:46171
S> WELCOME SimpleMail v1.0
C> MODE RECV
S> OK
S> AUTH REQUIRED RwQ8jPWy
C> AUTH alice 11170461993085822254
S> ERR Authentication failed
C> AUTH alice 11170461993085822254
S> ERR Authentication failed
C> AUTH alice 11170461993085822254
S> ERR Too many failures
[connection closed by server]
TCP connection: 127.0.0.1:46171
S> WELCOME SimpleMail v1.0
C> MODE RECV
S> OK
S> AUTH REQUIRED PVCswi9m
C> AUTH ALICE 90232564109656942 51
S> OK Welcome alice
C> COUNT
S> OK 0.
C> LIST
S> OK 0 messages
S> .
C> READ 999
S> ERR No such message
C> DELETE 999
S> ERR No such message
C> PING
S> ERR Unknown command.
C> QUIT
S> BYE

RESULT: PASS
```

Body limit and window overrun (1/7)

Ubuntu | Body limit and window overrun (1/7)

```
Ubuntu 24.04.3 LTS | Body-size boundary and credit overrun

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 60843 users.txt
TCP connection: 127.0.0.1:60843
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB 65536 bytes
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
```

Body limit and window overrun (2/7)

Ubuntu | Body limit and window overrun (2/7)

```
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
```

Body limit and window overrun (3/7)

Ubuntu | Body limit and window overrun (3/7)

```
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
```

Body limit and window overrun (4/7)

Ubuntu | Body limit and window overrun (4/7)

```
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
C> END_BODY
S> WINDOW_UPDATE 2048
S> OK Queued 1 to 1 mailboxes
stored body byte count: 65536
TCP connection: 127.0.0.1:60843
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB too large
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
```

Body limit and window overrun (5/7)

Ubuntu | Body limit and window overrun (5/7)

```
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
```

Body limit and window overrun (6/7)

Ubuntu | Body limit and window overrun (6/7)

```
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
```

Body limit and window overrun (7/7)

Ubuntu | Body limit and window overrun (7/7)

```
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
S> WINDOW_UPDATE 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
[65536 accepted body bytes; attempting one additional byte]
S> WINDOW_UPDATE 2048
C> CHUNK 1
S> ERR Body too large.
[connection closed by server]
TCP connection: 127.0.0.1:60843
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB credit overrun
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 2049
S> ERR Window overflow
[connection closed by server]

RESULT: PASS
```

TCP fragments and line length (1/2)

Ubuntu | TCP fragments and line length (1/2)

```
Ubuntu 24.04.3 LTS | Fragmented TCP input and command-length boundary

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 47135 users.txt
TCP connection: 127.0.0.1:47135
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Fragment Test
S> OK Sender accepted.
C> TO bob  [sent one byte at a time]
S> OK Recipient accepted.
C> SUB xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxx
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 21
C> [21 raw body bytes]
```

TCP fragments and line length (2/2)

Ubuntu | TCP fragments and line length (2/2)

```
S> OK Window 2027
C> END_BODY
S> WINDOW_UPDATE 21
S> OK Queued 1 to 1 mailboxes
TCP connection: 127.0.0.1:47135
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Long Command
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxx
S> ERR Invalid or overlong command line
[connection closed by server]

RESULT: PASS
```

Full queue, zero window and recovery (1/5)

Ubuntu | Full queue, zero window and recovery (1/5)

```
Ubuntu 24.04.3 LTS | Full ring, zero window and recovery

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 54275 users.txt
$ kill -STOP 1379
TCP connection: 127.0.0.1:54275
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 1 to 1 mailboxes
fill slot 1: OK Queued 1 to 1 mailboxes
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 2 to 1 mailboxes
fill slot 2: OK Queued 2 to 1 mailboxes
C> FROM Tarun Nagarajan
```

Full queue, zero window and recovery (2/5)

Ubuntu | Full queue, zero window and recovery (2/5)

```
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 3 to 1 mailboxes
fill slot 3: OK Queued 3 to 1 mailboxes
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 4 to 1 mailboxes
fill slot 4: OK Queued 4 to 1 mailboxes
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
```

Full queue, zero window and recovery (3/5)

Ubuntu | Full queue, zero window and recovery (3/5)

```
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 5 to 1 mailboxes
fill slot 5: OK Queued 5 to 1 mailboxes
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 6 to 1 mailboxes
fill slot 6: OK Queued 6 to 1 mailboxes
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 7 to 1 mailboxes
fill slot 7: OK Queued 7 to 1 mailboxes
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
```

Full queue, zero window and recovery (4/5)

Ubuntu | Full queue, zero window and recovery (4/5)

```
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 3
C> [3 raw body bytes]
S> OK Window 2045
C> END_BODY
S> WINDOW_UPDATE 3
S> OK Queued 8 to 1 mailboxes
fill slot 8: OK Queued 8 to 1 mailboxes
TCP connection: 127.0.0.1:54275
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB paused consumer
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
[window is zero; no credit update while ring is full]
TCP connection: 127.0.0.1:54275
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB ninth message
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> END_BODY
ninth END_BODY: no queue confirmation while all 8 slots are full
TCP connection: 127.0.0.1:54275
```

Full queue, zero window and recovery (5/5)

```
Ubuntu | Full queue, zero window and recovery (5/5)
```

```
S> WELCOME SimpleMail v1.0
C> MODE RECV
S> OK
S> AUTH REQUIRED feoCSIwb
C> AUTH bob 13450277825985278351
S> OK Welcome bob
C> COUNT
S> OK 0.
TCP connection: 127.0.0.1:54275
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB zero-credit violation
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 2048
C> [2048 raw body bytes]
S> OK Window 0
C> CHUNK 1
S> ERR Window overflow
[connection closed by server]
$ kill -CONT 1379
S> OK Queued 9 to 1 mailboxes
waiting sender: OK Queued 9 to 1 mailboxes
S> WINDOW_UPDATE 2048
C> CHUNK 6
C> [6 raw body bytes]
S> OK Window 2042
C> END_BODY
S> WINDOW_UPDATE 6
S> OK Queued 10 to 1 mailboxes
delivered mailbox count after recovery: 10

RESULT: PASS
```

Twenty concurrent senders (1/10)

```
Ubuntu | Twenty concurrent senders (1/10)
```

```
Ubuntu 24.04.3 LTS | Twenty simultaneous sending clients

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 52475 users.txt
=== Concurrent sender 01 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 7 to 1 mailboxes
sender 01: OK Queued 7 to 1 mailboxes
=== Concurrent sender 02 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
```

Twenty concurrent senders (2/10)

Ubuntu | Twenty concurrent senders (2/10)

```
S> OK Queued 12 to 1 mailboxes
sender 02: OK Queued 12 to 1 mailboxes
=== Concurrent sender 03 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> OK Queued 11 to 1 mailboxes
sender 03: OK Queued 11 to 1 mailboxes
=== Concurrent sender 04 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 10 to 1 mailboxes
sender 04: OK Queued 10 to 1 mailboxes
```

Twenty concurrent senders (3/10)

Ubuntu | Twenty concurrent senders (3/10)

```
=== Concurrent sender 05 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 2 to 1 mailboxes
sender 05: OK Queued 2 to 1 mailboxes
=== Concurrent sender 06 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 13 to 1 mailboxes
sender 06: OK Queued 13 to 1 mailboxes
=== Concurrent sender 07 ===
```

Twenty concurrent senders (4/10)

Ubuntu | Twenty concurrent senders (4/10)

```
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 14 to 1 mailboxes
sender 07: OK Queued 14 to 1 mailboxes
=== Concurrent sender 08 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 8 to 1 mailboxes
sender 08: OK Queued 8 to 1 mailboxes
=== Concurrent sender 09 ===
TCP connection: 127.0.0.1:52475
```

Twenty concurrent senders (5/10)

Ubuntu | Twenty concurrent senders (5/10)

```
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> WINDOW_UPDATE 12
S> OK Queued 4 to 1 mailboxes
sender 09: OK Queued 4 to 1 mailboxes
=== Concurrent sender 10 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 12
C> [12 raw body bytes]
S> OK Window 2036
C> END_BODY
S> OK Queued 15 to 1 mailboxes
sender 10: OK Queued 15 to 1 mailboxes
=== Concurrent sender 11 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
```

Twenty concurrent senders (6/10)

```
Ubuntu | Twenty concurrent senders (6/10)
```

```
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 17 to 1 mailboxes
sender 11: OK Queued 17 to 1 mailboxes
=== Concurrent sender 12 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> OK Queued 16 to 1 mailboxes
sender 12: OK Queued 16 to 1 mailboxes
=== Concurrent sender 13 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
```

Twenty concurrent senders (7/10)

```
Ubuntu | Twenty concurrent senders (7/10)
```

```
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 5 to 1 mailboxes
sender 13: OK Queued 5 to 1 mailboxes
=== Concurrent sender 14 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 1 to 1 mailboxes
sender 14: OK Queued 1 to 1 mailboxes
=== Concurrent sender 15 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
```

Twenty concurrent senders (8/10)

```
Ubuntu | Twenty concurrent senders (8/10)
```

```
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 18 to 1 mailboxes
sender 15: OK Queued 18 to 1 mailboxes
=== Concurrent sender 16 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> OK Queued 19 to 1 mailboxes
sender 16: OK Queued 19 to 1 mailboxes
=== Concurrent sender 17 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
```

Twenty concurrent senders (9/10)

```
Ubuntu | Twenty concurrent senders (9/10)
```

```
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 9 to 1 mailboxes
sender 17: OK Queued 9 to 1 mailboxes
=== Concurrent sender 18 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 6 to 1 mailboxes
sender 18: OK Queued 6 to 1 mailboxes
=== Concurrent sender 19 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
```

Twenty concurrent senders (10/10)

Ubuntu | Twenty concurrent senders (10/10)

```
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 3 to 1 mailboxes
sender 19: OK Queued 3 to 1 mailboxes
=== Concurrent sender 20 ===
TCP connection: 127.0.0.1:52475
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 13
C> [13 raw body bytes]
S> OK Window 2035
C> END_BODY
S> WINDOW_UPDATE 13
S> OK Queued 20 to 1 mailboxes
sender 20: OK Queued 20 to 1 mailboxes
20 simultaneous senders; 20 unique tickets; mailbox COUNT = 20
elapsed: 0.765 seconds

RESULT: PASS
```

IDs after deletion and restart (1/2)

Ubuntu | IDs after deletion and restart (1/2)

```
Ubuntu 24.04.3 LTS | Deletion and durable IDs across daemon restart

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 53069 users.txt
TCP connection: 127.0.0.1:53069
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 15
C> [15 raw body bytes]
S> OK Window 2033
C> END_BODY
S> WINDOW_UPDATE 15
S> OK Queued 1 to 1 mailboxes
TCP connection: 127.0.0.1:53069
S> WELCOME SimpleMail v1.0
C> MODE RECV
S> OK
S> AUTH REQUIRED UFBWKdUX
C> AUTH bob 13450277059698607213
S> OK Welcome bob
C> DELETE 1
S> OK Deleted.
C> COUNT
S> OK 0.
before restart .next-id: 2
$ kill -TERM 1404
$ kill -TERM 1403
```

IDs after deletion and restart (2/2)

Ubuntu | IDs after deletion and restart (2/2)

```
$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 52803 users.txt
TCP connection: 127.0.0.1:52803
S> WELCOME SimpleMail v1.0
C> MODE SEND
S> OK
C> FROM Tarun Nagarajan
S> OK Sender accepted.
C> TO bob
S> OK Recipient accepted.
C> SUB Demonstration
S> OK Subject accepted.
C> BODY_STREAM
S> OK Window 2048
C> CHUNK 15
C> [15 raw body bytes]
S> OK Window 2033
C> END_BODY
S> WINDOW_UPDATE 15
S> OK Queued 1 to 1 mailboxes
TCP connection: 127.0.0.1:52803
S> WELCOME SimpleMail v1.0
C> MODE RECV
S> OK
S> AUTH REQUIRED GcCYSdW0
C> AUTH bob 13450276500534771393
S> OK Welcome bob
C> LIST
S> OK 1 messages
S> 2    Tarun Nagarajan Demonstration   2026-10-29 10:39:05
S> .
mailbox files: 2.txt
mailbox ID 1 not reused; queue ticket sequence is scoped to a spooler run

RESULT: PASS
```

Interactive send menu

Ubuntu | Interactive send menu

```
Ubuntu 24.04.3 LTS | Interactive client

$ ./smclient 127.0.0.1 41699
1. Send a mail
2. Check my mailbox
3. Quit
Choice: 1
WELCOME SimpleMail v1.0
OK
Sender display name: Tarun Nagarajan
OK Sender accepted.
Recipient username (blank when finished): bob
OK Recipient accepted.
Recipient username (blank when finished):
Subject (blank for no subject): Client menu test
OK Subject accepted.
Enter the body one line at a time. A line containing only a single period ends it.
Client output from Ubuntu
.
OK Window 2048
OK Window 2021
Server credit update: WINDOW_UPDATE 27
OK Queued 1 to 1 mailboxes
3
1. Send a mail
2. Check my mailbox
3. Quit
Choice:
```

Interactive mailbox menu

Ubuntu | Interactive mailbox menu

```
$ ./smclient 127.0.0.1 41699
1. Send a mail
2. Check my mailbox
3. Quit
Choice: 2
WELCOME SimpleMail v1.0
Mailbox nonce: 1xz6y3o4
Username: bob
Password: secret2
OK Welcome bob
Mailbox: [L]ist, [R]ead, [D]elete, [C]ount, [Q]uit
Choice: L
OK 1 messages
1   Tarun Nagarajan Client menu test    2026-10-29 10:39:07
.
Mailbox: [L]ist, [R]ead, [D]elete, [C]ount, [Q]uit
Choice: R
Message ID: 1
OK
From: Tarun Nagarajan
To: bob
Subject: Client menu test
Date: 2026-10-29 10:39:07
---

Client output from Ubuntu
Mailbox: [L]ist, [R]ead, [D]elete, [C]ount, [Q]uit
Choice: D
Message ID: 1
OK Deleted.
Mailbox: [L]ist, [R]ead, [D]elete, [C]ount, [Q]uit
Choice: C
OK 0.
Mailbox: [L]ist, [R]ead, [D]elete, [C]ount, [Q]uit
Choice: Q
BYE
1. Send a mail
2. Check my mailbox
3. Quit
Choice: 3

RESULT: PASS
```

Timestamped server and spooler events

Ubuntu | Timestamped server and spooler events

```
Ubuntu 24.04.3 LTS | Server and spooler events

$ /mnt/c/Users/ultim/SimpleMail/smspooler users.txt
$ /mnt/c/Users/ultim/SimpleMail/smserver 41699 users.txt
[2026-10-29 10:39:07] server: listening on port 41699 with 3 users
[2026-10-29 10:39:07] server: client connected
[2026-10-29 10:39:07] server: selected SEND mode
[2026-10-29 10:39:07] server: queued ticket 1 from Tarun Nagarajan to 1 mailboxes
[2026-10-29 10:39:07] server: client disconnected
[2026-10-29 10:39:07] server: client connected
[2026-10-29 10:39:07] server: selected RECV mode
[2026-10-29 10:39:07] server: authenticated mailbox bob
[2026-10-29 10:39:07] server: mailbox bob counted
[2026-10-29 10:39:07] server: client disconnected
[2026-10-29 10:39:07] server: client connected
[2026-10-29 10:39:07] server: selected RECV mode
[2026-10-29 10:39:07] server: authenticated mailbox bob
[2026-10-29 10:39:07] server: mailbox bob listed
[2026-10-29 10:39:07] server: mailbox bob read message 1
[2026-10-29 10:39:07] server: mailbox bob deleted message 1
[2026-10-29 10:39:07] server: mailbox bob counted
[2026-10-29 10:39:07] server: mailbox bob logged out
[2026-10-29 10:39:07] server: client disconnected
[2026-10-29 10:39:07] server: stopping
[2026-10-29 10:39:07] spooler: ready; 3 registered users, queue capacity 8
[2026-10-29 10:39:07] spooler: ticket 1 dequeued (1 recipients)
[2026-10-29 10:39:07] spooler: ticket 1 delivered to bob
[2026-10-29 10:39:08] spooler: stopping

RESULT: PASS
```